Test Plan – Triangle and Rectangle Features

Project Name: Polygon Checker
Issue: #7 – Update Test Plan for Rectangle (Second Approach)
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1. Test Objectives and Scope

The project has created a plan that introduces ways to test out the two main features of the Polygon Checker program which are the triangle and the rectangle.
Starting with the triangle feature, the program's capability to accurately recognize valid triangles, categorize them and measure the interior angles precisely is the test case. Triangles which are invalid or impossible should be properly rejected by the program without any crashing.
The second method shows the rectangle feature, where the objective is to validate that the system is able to recognize the corresponding four points, given in any random order, as a proper rectangle. The system should automatically arrange the points to set up corners, restrict lines from crossing, and compute the perimeter and area correctly. It should also take care of wrong inputs like duplicate or collinear points smoothly.

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2. Rectangle Feature Test Cases

The following cases will be tested for the rectangle feature:

1. **R1 – Valid Rectangle in Random Order:**
   Input points: (3,2), (0,2), (0,0), (3,0).
   Expected result: The program should recognize the shape as a rectangle with an area of 6 and a perimeter of 10.

2. **R2 – Invalid Parallelogram:**
   Input points: (0,0), (1,2), (3,2), (2,0).
   Expected result: The program should state that the shape is *not a rectangle*.

3. **R3 – Square (Special Case):**
   Input points: (0,0), (0,2), (2,2), (2,0).
   Expected result: The program should recognize it as a rectangle with an area of 4 and a

perimeter of 8.

4. **R4 – Duplicate Points:**
   Input points: (0,0), (0,0), (2,2), (2,0).
   Expected result: The program should detect the duplicate points and show an *invalid input* error.

5. **R5 – Collinear Points:**
   Input points: (0,0), (1,1), (2,2), (3,3).
   Expected result: The program should correctly identify that the points are collinear and cannot form a rectangle.

6. **R6 – Random Order Rectangle:**
   Input points: (2,0), (0,2), (2,2), (0,0).
   Expected result: The system should identify this as a rectangle with an area of 4 and a perimeter of 8.

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3. Rectangle Validation Results Summary

All six rectangle test cases were executed successfully.
Each case produced the expected output:

- R1: Passed — recognized rectangle correctly.
- R2: Passed — correctly rejected a parallelogram.
- R3: Passed — square correctly handled as a rectangle.
- R4: Passed — duplicate points detected and rejected.
- R5: Passed — collinear points rejected.
- R6: Passed — correctly identified rectangle regardless of input order.

Overall, the rectangle feature passed all validation tests. The function correctly determines rectangle geometry even when the user enters points in any random order.

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4. Notes and References

- The **rectangle geometry code** was implemented in Issue #6.
- The **unit tests** for the rectangle feature were completed in Issue #5.
- This test plan is part of the **second approach**, where the program automatically determines point order.
- The updated version of this document includes both **triangle** and **rectangle** test cases.